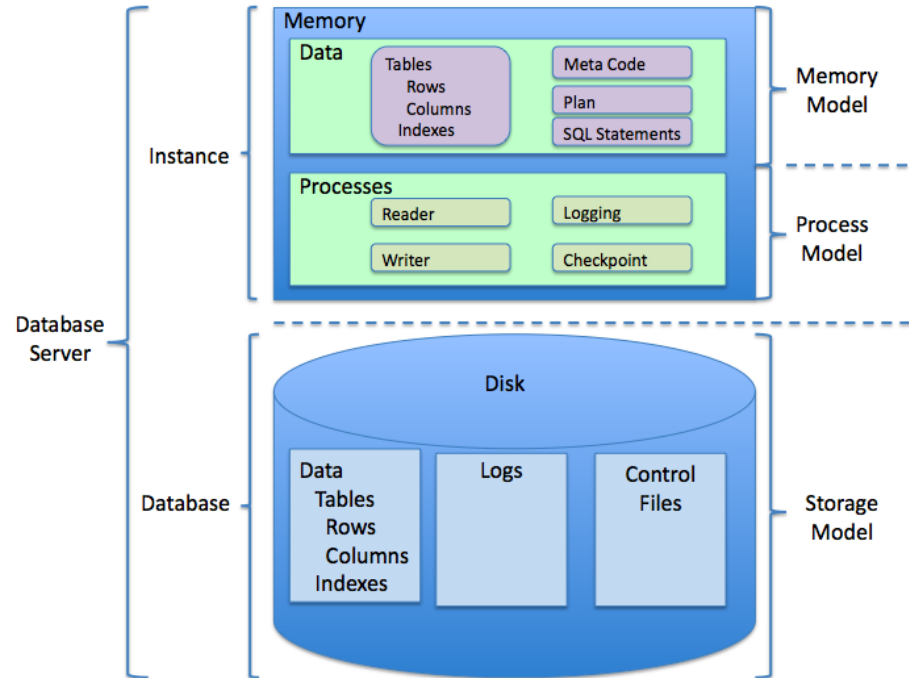


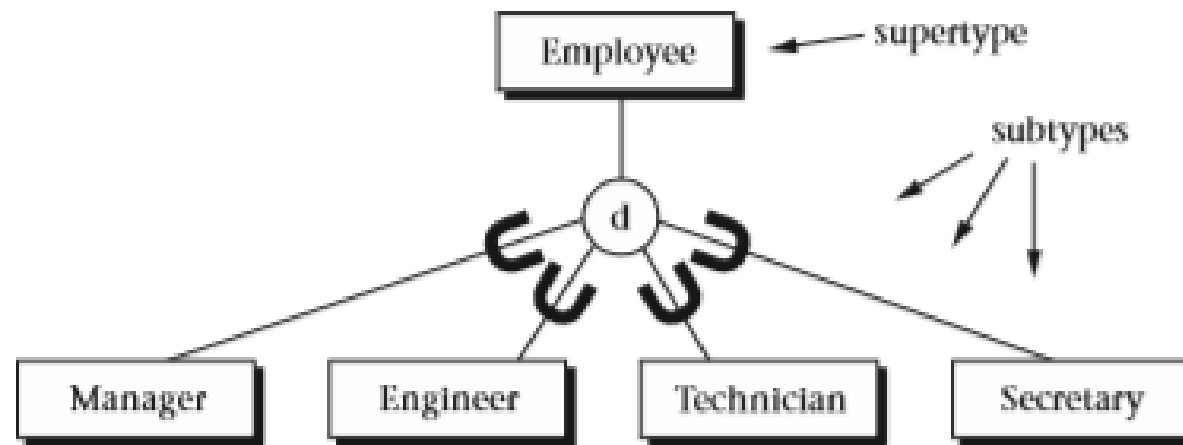
# Database Management System



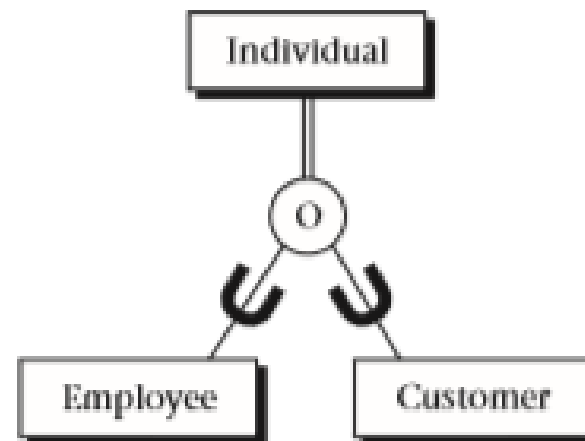
By Assistant Professor Malik M Ali

# Advance ER Constructs

- **Generalization:** The generalization relationship specifies that several types of entities with certain common attributes can be generalized into a higher-level entity type—a generic or superclass entity, more commonly known as a super-type entity.



(a) Generalization with disjoint subtypes

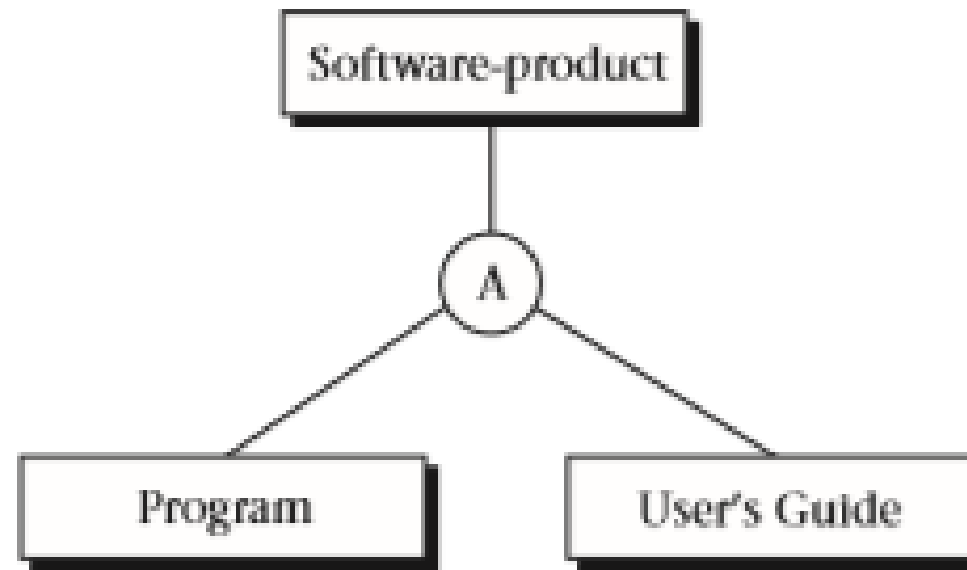


(b) Generalization with overlapping subtypes and completeness constraint

# Generalization – completeness

- The completeness constraint requires the subtypes to be all-inclusive of the supertype. Thus, subtypes can be defined as either total or partial coverage of the supertype. For example, in a generalization hierarchy with supertype **Individual** and subtypes **Employee** and **Customer**, the subtypes may be described as all-inclusive or total. We denote this type of constraint by a double line between the supertype entity and the circle.

# Aggregation



# Aggregation

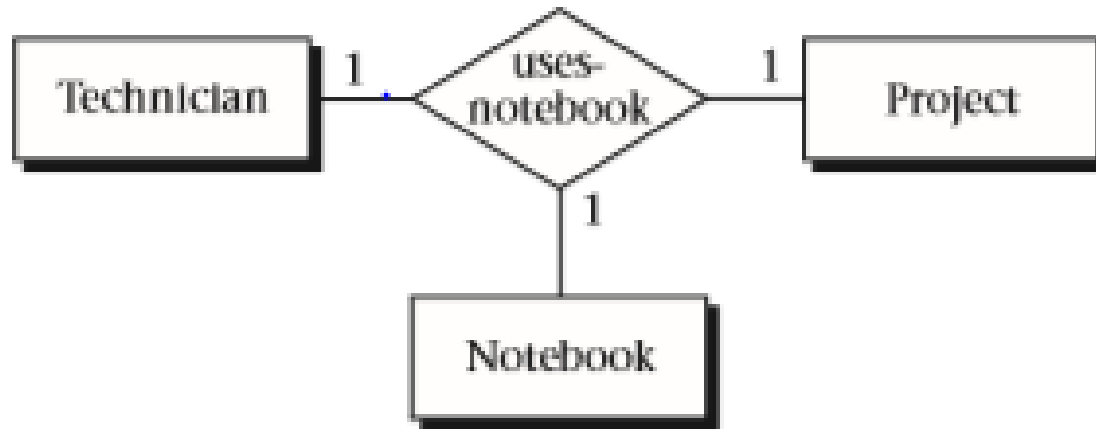
- Aggregation is a form of abstraction between a supertype and subtype entity.
- Generalization is often described in terms of an “is-a” relationship between the subtype and the supertype. For example, an Employee is an Individual.
- Aggregation, on the other hand, is the relationship between the whole and its parts and is described as a “part-of” relationship. For example, a report and a prototype software package are both parts of a deliverable for a contract

# Ternary Relationships

- It is a relationship that define semantics of an association among three entities.
- An entity in a ternary relationship is considered to be “one” if only one instance of it can be associated with one instance of each of the other two associated entities.
- It is “many” if more than one instance of it can be associated with one instance of each of the other two associated entities.

# Ternary Relationship – Example 01

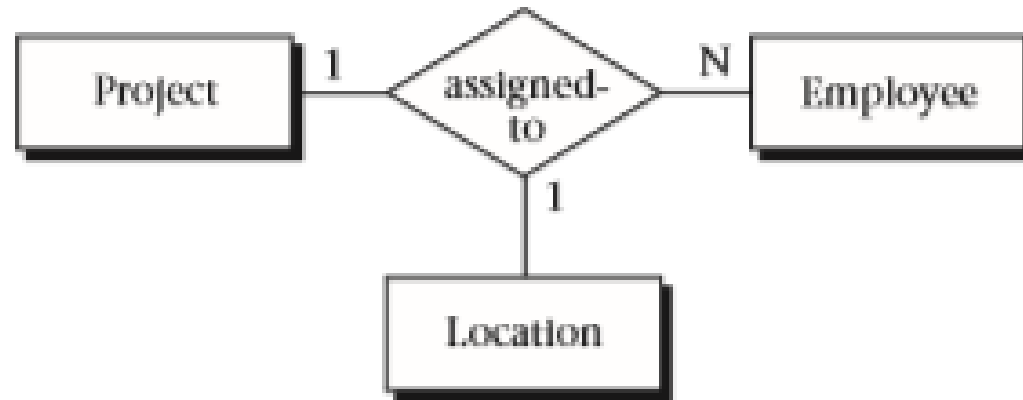
- A technician uses exactly one notebook for each project. Each notebook belongs to one technician for each project. Note that a technician may still work on many projects and maintain different notebooks for different projects





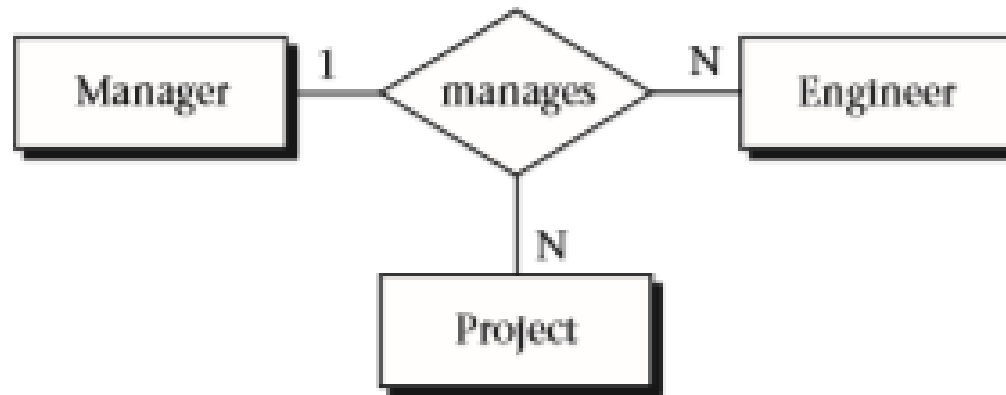
# Ternary Relationship – Example 02

- Each employee assigned to a project works at only one location for that project, but can be at different locations for different projects. At a particular location, an employee works on only one project. At a particular location, there can be many employees assigned to a given project.



# Ternary Relationship – Example 03

- Each engineer working on a particular project has exactly one manager, but each manager of a project may manage many engineers, and each manager of an engineer may manage that engineer on many projects.

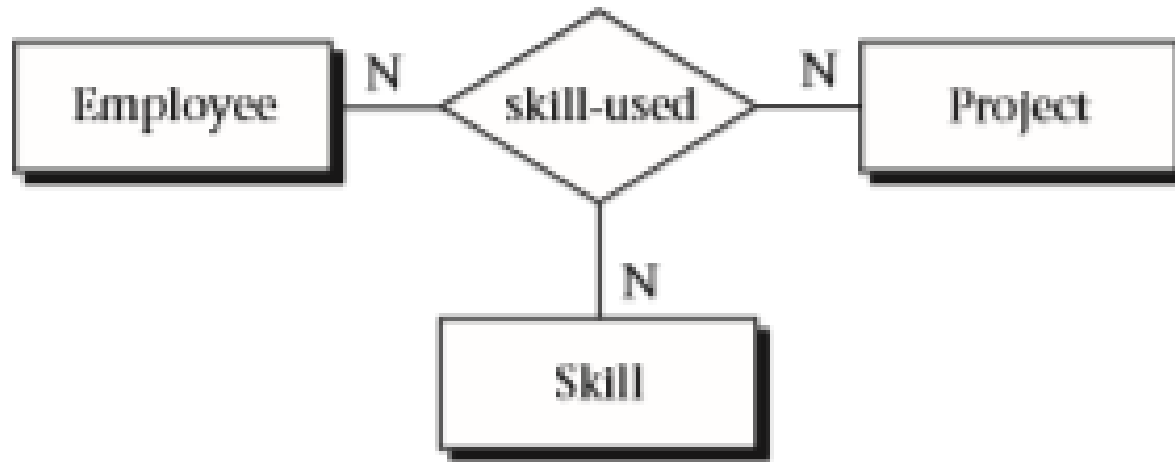


# Some explanation about Example 03

- The manager is considered as ONE because of the following assertions
  - Assertion 1: One engineer, working under one manager, could be working on many projects.
  - Assertion 2: One project, under the direction of one manager, could have many engineers.
  - Assertion 3: One engineer, working on one project, must have only a single manager

# Ternary Relationship – Example 04

- Employees can use many skills on any one of many projects, and each project has many employees with various skills



# General n-ary Relationships

- Generalizing the ternary form to higher-degree relationships, an n-ary relationship that describes some association among n entities is represented by a single relationship diamond with n connections

